



**UNIVERSITY OF PATRAS**  
**COMPUTER ENGINEERING AND INFORMATICS DEPARTMENT**  
**MSc PROGRAM *DATA DRIVEN COMPUTING AND DECISION MAKING***

MSc topics 2021-2022  
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## **1. Community discovery in multilayer social networks**

Community discovery is a major problem in social network analysis as they reveal strong similarities among accounts such as common interests or activities. As today people are likely to be connected on multiple social media, mining these parallel connections or the absence thereof may reveal considerable more patterns than by mining each connection set alone. For instance, LinkedIn interactions are primarily of professional nature, whereas Twitter interactions may pertain to various topics with attributes like hashtags and frequent words providing context. In this MSc thesis data will be collected from at least two social networks through their respective APIs for a large group of accounts, their interactions will be modeled as a third order tensor, and communities will be derived.

## **2. Tensor algebra taxonomy**

Tensor algebra is emerging as an indispensable tool for pattern mining and anomaly discovery in multilayer graphs and other types of inherently multidimensional data. Applications include multispectral imaging, financial engineering, and social network analysis. The primary research objective is the construction of a taxonomy encompassing major algorithmic techniques such as multilinear systems solvers, eigentensor computation, and tensor transforms aiming at solving social network analysis problems such as community discovery. As a secondary objective, conditions for employing each algorithm like the computational cost, the memory requirements, and the numerical stability will be derived. The above will be tested on a benchmark dataset using Julia.